Solutions - Continuous Probability distributions

1. Let x = debt amount

$\mu = 15{,}015, \quad \sigma = 3540$

a. $z = \dfrac{18{,}000 - 15{,}015}{3540} = .84$

$P(x > 18{,}000) = 1 - P(z \le .84) = 1 - .7995 = .2005$

b. $z = \dfrac{10{,}000 - 15{,}015}{3540} = -1.42$

$P(x < 10{,}000) = P(z < -1.42) = .0778$

c. At 18,000, $z = .84$ from part (a)

At 12,000, $z = \dfrac{12{,}000 - 15{,}015}{3540} = -.85$

$P(12{,}000 < x < 18{,}000) = P(-.85 < z < .84) = .7995 - .1977 = .6018$

d. $z = \dfrac{14{,}000 - 15{,}015}{3540} = -.29$

$P(x \le 14{,}000) = P(z \le -.29) = .3859$

2. $\mu = 14.4$ and $\sigma = 4.4$

a. At $x = 20$, $z = \dfrac{20 - 14.4}{4.4} = 1.27$

$P(z \le 1.27) = .8980$

$P(x \ge 20) = 1 - .8980 = .1020$

b. At $x = 10$, $z = \dfrac{10 - 14.4}{4.4} = -1.00$

$P(z \le -1.00) = .1587$

So, $P(x \le 10) = .1587$

c. A z-value of 1.28 cuts off an area of approximately 10% in the upper tail.

$x = 14.4 + 4.4(1.28) = 20.03$

A return of 20.03% or higher will put a domestic stock fund in the top 10%

3. $\mu = 328$ and $\sigma = 92$

a. $z = \dfrac{x - \mu}{\sigma} = \dfrac{500 - 328}{92} = 1.87$

$P(x > 500) = P(z > 1.87) = 1 - P(z \le 1.87) = 1 - .9693 = .0307$

The probability that the emergency room visit will cost more than $500 is .0307.

b. $z = \dfrac{x - \mu}{\sigma} = \dfrac{250 - 328}{92} = -.85$

$P(x < 250) = P(z < -.85) = .1977$

The probability that the emergency room visit will cost less than $250 is .1977.

c. For $x = 400$, $z = \frac{x-\mu}{\sigma} = \frac{400-328}{92} = .78$

For $x = 300$, $z = \frac{x-\mu}{\sigma} = \frac{300-328}{92} = -.30$

$P(300 < x < 400) = P(z \leq .78) - P(z \leq -.30) = .7823 - .3821 = .4002$

The probability that the emergency room visit will cost between \$300 and \$400 is .4002.

d. The lower 8%, or area = .08, occurs for $z = -1.41$

$x = \mu + z\sigma = 328 - 1.41(92) = \$198.28$

For a patient to have a charge in the lower 8%, the cost of the visit must have been \$198.28 or less.

4. Use $\mu = 8.35$ and $\sigma = 2.5$

a. We want to find $P(5 \leq x \leq 10)$

At $x = 10$,

$$z = \frac{x-\mu}{\sigma} = \frac{10-8.35}{2.5} = .66$$

At $x = 5$,

$$z = \frac{x-\mu}{\sigma} = \frac{5-8.35}{2.5} = -1.34$$

$P(5 \leq x \leq 10) = P(-1.34 \leq z \leq .66) = P(z \leq .66) - P(z \leq -1.34)$

$= .7454 - .0901$

= .6553

The probability of a household viewing television between 5 and 10 hours a day is .6553.

b. Find the z-value that cuts off an area of .03 in the upper tail. Using a cumulative probability of 1 - .03 = .97, z = 1.88 provides an area of .03 in the upper tail of the normal distribution.

$$x = \mu + z\sigma = 8.35 + 1.88(2.5) = 13.05 \text{ hours}$$

A household must view slightly over 13 hours of television a day to be in the top 3% of television viewing households.

c. At $x = 3$, $z = \frac{x - \mu}{\sigma} = \frac{3 - 8.35}{2.5} = -2.14$

$P(x>3) = 1 - P(z\leq -2.14) = 1 - .0162 = .9838$

The probability a household views more than 3 hours of television a day is .9838.

5. $\mu = 6.8, \quad \sigma = .6$

a. At $x = 8$, $z = \frac{8 - 6.8}{.6} = 2.00$

$P(x > 8) = P(z > 2.0) = 1 - .9772 = .0228$

b. At $x = 6$, $z = \frac{6 - 6.8}{.6} = -1.33$

$P(x \leq 6) = P(z \leq -1.33) = .0918$

c. At $x = 9$, $z = \frac{9 - 6.8}{.6} = 3.67$

At $x = 7,\ z = \dfrac{7 - 6.8}{.6} = .33$

$P(7 < x < 9) = P(.33 < z < 3.67) = 1 - .6293 = .3707$

Only 37.07 percent of the population gets the amount of sleep recommended by doctors. Most get less.

6. a. United States: $\mu = 3.73 \quad \sigma = .25$

At $x = 3.50,\ z = \dfrac{3.5 - 3.73}{.25} = -.92$

$P(z < -.92) = .1788$

So, $P(x < 3.50) = .1788$

b. Russia: $\mu = 3.40 \quad \sigma = .20$

At $x = 3.50,\ z = \dfrac{3.50 - 3.40}{.20} = .50$

$P(z < .50) = .6915$

So, $P(x < 3.50) = .6915$

69.15% of the gas stations in Russia charge less than $3.50 per gallon.

c. Use mean and standard deviation for Russia.

At $x = 3.73,\ z = \dfrac{3.73 - 3.40}{.20} = 1.65$

$P(z > 1.65) = 1 - P(z \le 1.65) = 1 - .9505 = .0495$

$P(x > 3.73) = .0495$

The probability that a randomly selected gas station in Russia charges more than the mean price in the United States is .0495. Stated another way, only 4.95% of the gas stations in Russia charge more than the average price in the United States.

7. a. $z = \frac{60-80}{10} = -2$ $P(z \le -2) = .0228$. So $P(x < 60) = .0228$

b. At $x = 60$

$$z = \frac{60-80}{10} = -2$$ Area to left is .0228

At $x = 75$

$$z = \frac{75-80}{10} = -.5$$ Area to left is .3085

$P(60 \le x \le 75) = .3085 - .0228 = .2857$

c. $z = \frac{90-80}{10} = 1$ $P(z \le 1) = P(x \le 90) = .1587$

Therefore 15.87% of students will not complete on time.

$$(60)\ (.1587) = 9.52$$

We would expect 9 or 10 students to be unable to complete the exam in time.

8. a. $\bar{x} = \sum \frac{x_i}{n} = 200$

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 26.04$$

We will use $\bar{x}$ as an estimate of μ and s as an estimate of σ in parts (b) - (d) below.

b. Remember the data are in thousands of shares.

At $x = 180$

$$z = \frac{180 - 200}{26.04} = -.77$$

$$P(x \leq 180) = P(z \leq -.77) = .2206$$

The probability trading volume will be less than 180 million shares is .2206.

c. At $x = 230$

$$z = \frac{230 - 200}{26.04} = 1.15$$

$$P(x > 230) = P(z > 1.15) = 1 - P(z \leq 1.15) = 1 - .8749 = .1251$$

The probability trading volume will exceed 230 million shares is .1251.

d. A z-value of 1.645 cuts off an area of .05 in the upper tail

$$x = \mu + z\sigma = 200 + 1.645(26.04) = 242.84$$

If the early morning trading volume exceeds 242.84 million shares, the day is among the busiest 5%.